S181M116 Derivatives and Alternative Investments

Lecture 2: Hedging with Futures and Forward/Futures Prices

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Today

1. How to choose a short hedge or a long hedge.
2. Why hedges remain imperfect because of basis risk.
3. How to size cross hedges.
4. How no-arbitrage determines forward and futures prices.
5. Why commodity pricing needs storage costs and convenience yield.
6. How stock-index futures can hedge or change portfolio beta.
7. Why stack-and-roll hedging creates roll and funding risk.

Core Message

Lecture 2 has two linked problems

1. **Hedging:** choose a futures position that offsets an exposure.
2. **Pricing:** explain why spot and futures prices should move together.

A hedge does not make the firm win in every state. It trades upside for downside protection, and the remaining gap is usually basis risk.

Section 1

Hedge Mechanics

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. Commodity Behavior
5. Stock Index Futures and Beta Hedging
6. Stack-and-Roll Hedging

Exposure Comes First

Before choosing a futures position, ask:

- Will the firm **sell** the asset later or **buy** it later?
- Is the futures contract written on the **same asset**, or only on a **related asset**?
- Do quantity, timing, location, and currency match the exposure?

If it is only a related asset, the firm is using a **cross hedge**, so the hedge will be less exact.

Short hedge

Use a short futures position when the firm owns, produces, or expects to receive an asset that it will later sell.

Long hedge

Use a long futures position when the firm expects to buy an asset later and wants protection against rising prices.

Short Hedge Example

An oil producer will sell 500,000 barrels in three months. The futures price is USD 78 per barrel and each contract is 1,000 barrels, so

$$N = \frac{500{,}000}{1{,}000} = 500$$

short contracts are needed.

If spot and futures finish near 72

- Physical sale: USD 36.0m
- Futures gain: USD 3.0m
- Total revenue: USD 39.0m

If spot and futures finish near 84

- Physical sale: USD 42.0m
- Futures loss: USD 3.0m
- Total revenue: USD 39.0m

Lesson

Ignoring basis risk, the short hedge locks in approximately the initial futures price.

Long Hedge Example

A manufacturer will buy 250,000 pounds of copper in four months. The futures price is USD 3.70 per pound and each contract covers 25,000 pounds, so

$$N = \frac{250{,}000}{25{,}000} = 10$$

long contracts are needed.

If copper rises to 4.10

- Physical purchase: USD 1.025m
- Futures gain: USD 0.100m
- Net cost: USD 0.925m

If copper falls to 3.40

- Physical purchase is cheaper.
- The futures position loses.
- Net cost is still close to USD 3.70 per pound.

Rule of Thumb

If the firm's exposure moves roughly **one-for-one** with the asset price, a **fixed number of futures contracts** often works well. If the exposure itself changes with price or time, a fixed futures hedge is usually not enough, so use **options** or **dynamic rebalancing**.

Why Firms Hedge

Economic reasons

- Reduce input-cost or revenue volatility.
- Improve budgeting and investment planning.
- Hedge more cheaply than many individual shareholders can.
- Reduce distress and funding risk from volatile cash flow.

Common mistake

Judge the **combined** business-plus-futures outcome. A hedge can lose money on the futures account precisely because it protected the underlying business exposure.

Hedging can even be counterproductive if the firm's margins already adjust naturally with input prices.

Section 2

Basis Risk

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. Commodity Behavior
5. Stock Index Futures and Beta Hedging
6. Stack-and-Roll Hedging
7. Wrap-Up

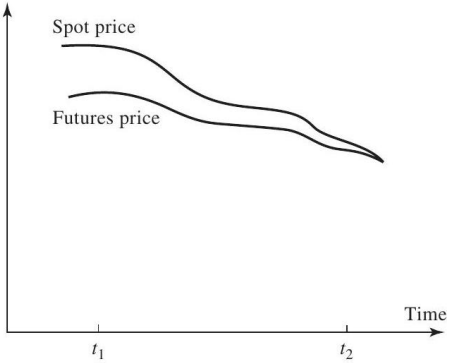
Basis Risk

Definition

$$b_t = S_t - F_t$$

where basis means **spot minus futures**.

- Spot and futures prices usually move together.
- For the same deliverable asset, basis should be close to zero at true futures maturity.
- But firms often close hedges before maturity, or hedge with an imperfect match.
- So the basis at **hedge close** can still be nonzero; that uncertainty is **basis risk**.



Effective Sale Price in a Short Hedge

Cash-flow logic

Sell the physical asset later at spot price S_2.
Short futures gain:

$$F_1 - F_2$$

Effective sale price:

$$S_2 + (F_1 - F_2) = F_1 + b_2$$

where $b_2 = S_2 - F_2$.

Example

$$\underbrace{5.85}_{\text{spot sale}} + \underbrace{(6.20 - 5.95)}_{\text{futures gain}} = 6.10$$

Intuition

- The firm does **not** pre-sell the physical asset at F_1.
- It still sells later at the market price S_2.
- The short futures position adds cash when prices fall.
- If $b_2 = 0$, the effective sale price is exactly F_1; otherwise basis risk is the remaining gap.

Effective Purchase Price in a Long Hedge

Cash-flow logic

Buy the physical asset later at spot price S_2.
Long futures gain:

$$F_2 - F_1$$

Effective purchase price:

$$S_2 - (F_2 - F_1) = F_1 + b_2$$

where $b_2 = S_2 - F_2$.

Example

$$\underbrace{3.92}_{\text{spot purchase}} - \underbrace{(3.88 - 3.70)}_{\text{futures gain}} = 3.74$$

Intuition

- The firm does **not** buy the physical asset today at F_1.
- It still buys later at the market price S_2.
- The long futures position reimburses part of the purchase when prices rise.
- If $b_2 = 0$, the effective purchase price is exactly F_1; otherwise basis determines whether it ends above or below F_1.

Cross Hedging

When is a cross hedge used?

When there is no liquid futures contract on the exact exposure, hedge with a related asset instead.

Example

An airline may hedge jet fuel with heating-oil futures if jet-fuel futures are illiquid.

$$S_2 - F_2 = (S_2 - S_2^*) + (S_2^* - F_2)$$

- The first term is the mismatch between the two physical assets.
- The second term is the usual basis for the futures underlying.
- Cross hedges therefore add one more source of imperfection.

Minimum-Variance Hedge Ratio

If a short hedge uses ratio h, the change in the hedged position is

$$\Delta V = \Delta S + h(-\Delta F) = \Delta S - h\Delta F,$$

because a short futures position changes by $-\Delta F$: it offsets the spot exposure instead of adding to it. Choose h to make the residual risk as small as possible:

$$\mathrm{Var}(\Delta V) = \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F).$$

Optimal hedge ratio

$$h^* = \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)} = \rho\frac{\sigma_S}{\sigma_F}.$$

Full derivation in appendix

Intuition

- Small h: under-hedged.
- Large h: futures add noise.
- h^* minimizes residual variance.

Regression view

In $\Delta S = a + h^*\Delta F + \varepsilon$, the slope is the best-fit hedge ratio.

From Hedge Ratio to Contracts

Contract count

$$N^* = h^* \frac{Q_A}{Q_F} \qquad \text{or} \qquad N^* = h^* \frac{V_A}{V_F}.$$

Jet-fuel cross hedge

If $\rho = 0.80$, $\sigma_S = 0.18$, and $\sigma_F = 0.15$, then

$$h^* = 0.80 \frac{0.18}{0.15} = 0.96.$$

Hedging 3,000,000 gallons with 42,000-gallon contracts gives

$$N^* = 0.96 \frac{3{,}000{,}000}{42{,}000} \approx 68.6,$$

so the airline uses about 68 or 69 contracts.

If the regression uses percentage returns, base the contract count on values rather than physical quantities.

Section 3

Forward and Futures Pricing

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. Commodity Behavior
5. Stock Index Futures and Beta Hedging
6. Stack-and-Roll Hedging

Investment Assets, Consumption Assets, and No-Arbitrage

Investment assets

Stocks, bonds, gold, and silver are widely held for investment. Cash-and-carry arbitrage often pins down the forward price tightly.

Consumption assets

Oil, copper, gas, wheat, and electricity are held for use. Physical ownership can provide convenience benefits that a futures contract does not.

For a non-income-paying investment asset,

$$F_0 = S_0 e^{rT}.$$

Benchmark assumptions: no transaction costs, same borrowing and lending rate, short selling when needed, known income or storage costs, and no default risk.

Cash-and-Carry Arbitrage

Suppose $S_0 = 80$, $r = 4\%$ continuously compounded, and $T = 0.75$. Then the fair forward price is

$$F_0 = 80e^{0.04 \times 0.75} = 80e^{0.03} \approx 82.44.$$

If the quoted forward is 85

1. Borrow 80.
2. Buy the stock.
3. Short the forward.

At maturity, profit is

$$85 - 82.44 = 2.56.$$

If the quoted forward is 81

1. Short the stock.
2. Invest the proceeds.
3. Go long the forward.

At maturity, profit is

$$82.44 - 81 = 1.44.$$

Income and Yield in Forward Pricing

Known cash income

$$F_0 = (S_0 - I)e^{rT}$$

Example: $S_0 = 100$, known dividend PV $I \approx 2.93$, $r = 5\%$, $T = 1$:

$$F_0 = (100 - 2.93)e^{0.05} \approx 102.05.$$

Known yield

$$F_0 = S_0 e^{(r-q)T}$$

Useful for stock indices and foreign currencies, where holding the asset gives a running benefit.

Value of an Existing Forward Contract

Old versus new: An old forward contract has delivery price K fixed in the past. A new forward entered today would have delivery price F_0 and zero value today.

At maturity:

$$\text{old long payoff} = S_T - K, \qquad \text{new long payoff} = S_T - F_0.$$

So the old contract is better by

$$(S_T - K) - (S_T - F_0) = F_0 - K.$$

This difference is known today but paid at maturity, so its value today is

$$f = (F_0 - K)e^{-rT}.$$

Example: if your old long forward lets you buy at 50 while a new forward today would lock in 54, then your contract is favorable:

$$f = (54 - 50)e^{-0.04\times 0.5} \approx 3.92.$$

Commodity Pricing: Storage and Convenience Yield

Storage cost

Fixed storage cost PV U:

$$F_0 = (S_0 + U)e^{rT}$$

Proportional storage cost u:

$$F_0 = S_0 e^{(r+u)T}$$

Convenience yield

Physical inventory can keep production running or protect against shortages. With proportional storage cost u and convenience yield y:

$$F_0 = S_0 e^{(r+u-y)T}.$$

Oil example

If $S_0 = 80$, $r = 5\%$, $u = 1\%$, $y = 4\%$, and $T = 0.5$,

$$F_0 = 80e^{(0.05+0.01-0.04)(0.5)} \approx 80.80.$$

Without convenience yield, the price would be about 82.44.

Commodity Curves and Arbitrage Bounds

Investment commodity

Gold or silver can often be priced closely by arbitrage from spot, financing, and storage costs.

Consumption commodity

For oil or copper, arbitrage may give only an upper bound:

$$F_0 \leq (S_0 + U)e^{rT}.$$

A low futures price is not automatically an arbitrage opportunity because inventory has convenience value.

Curve shape

Contango: futures prices rise with maturity.

Backwardation: futures prices fall with maturity.

- Contango often reflects financing and storage costs.
- Backwardation often reflects scarcity or high convenience yield.

Futures, Forwards, and Expected Spot Prices

Futures versus forwards

They are often close, but daily settlement matters. If futures gains arrive when interest rates are high, futures prices can exceed forward prices, and vice versa.

Futures price is not just a forecast

$$F_0 = \mathbb{E}(S_T)e^{(r-k)T}$$

where k is the required expected return on the underlying asset.

Systematic risk	Relation	Implication
None	$k = r$	$F_0 = \mathbb{E}(S_T)$
Positive	$k > r$	$F_0 < \mathbb{E}(S_T)$
Negative	$k < r$	$F_0 > \mathbb{E}(S_T)$

Do not confuse **backwardation/contango** as curve shapes with **normal backwardation/contango** as statements about expected future spot prices.

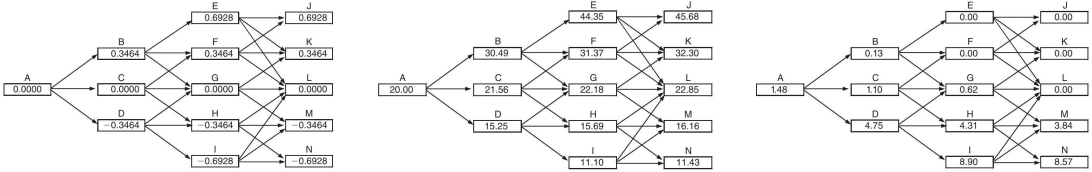
Section 4

Commodity Behavior

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. **Commodity Behavior**
5. Stock Index Futures and Beta Hedging
6. Stack-and-Roll Hedging

Commodity Markets Behave Differently

- **Agriculture:** seasonality, weather shocks, storage limits, and harvest uncertainty matter.
- **Metals:** gold and silver look more like investment assets; copper looks more like a production input.
- **Energy:** crude oil is storable, natural gas is highly seasonal, and electricity can spike because it is hard to store.
- **Modeling implication:** commodity models often need mean reversion, seasonality, jumps, storage costs, and convenience yields.



Section 5

Stock Index Futures and Beta Hedging

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. Commodity Behavior
5. **Stock Index Futures and Beta Hedging**
6. Stack-and-Roll Hedging

Intuition for Beta Hedging

Core idea

A stock portfolio has **market exposure** and **stock-specific exposure**. Index futures mainly hedge the **market-wide** part.

- Beta tells how strongly the portfolio tends to move with the index.
- A USD 40m portfolio with beta 1.25 behaves like about USD 50m of market exposure.
- If one futures contract gives USD 1m of index exposure, shorting 50 contracts offsets that market exposure.
- If the market falls, the portfolio loses but the short futures gain; if the market rises, the opposite happens.
- The manager can reduce market risk quickly without selling the underlying stocks.

What the hedge does not remove

Stock selection risk, sector tilts, changing beta, and futures basis risk can still remain.

Stock-Index Futures and Beta Hedging

One index futures contract has value

$$V_F = \text{futures price} \times \text{multiplier}.$$

If a portfolio has beta β relative to the index, the number of contracts to short is

$$N^* = \beta \frac{V_A}{V_F}.$$

Full market hedge

Portfolio value: USD 40m, beta: 1.25, index futures: 4,000, multiplier: 250. Then

$$V_F = 4{,}000 \times 250 = 1{,}000{,}000, \qquad N^* = 1.25 \frac{40{,}000{,}000}{1{,}000{,}000} = 50.$$

Short 50 contracts.

Changing Portfolio Beta

Futures can change exposure without selling the underlying stocks:

$$N^* = (\beta - \beta^*)\frac{V_A}{V_F}.$$

Reduce beta

From 1.40 to 0.80 on a USD 25m portfolio with USD 250,000 per contract:

$$N^* = (1.40 - 0.80)\frac{25{,}000{,}000}{250{,}000} = 60.$$

Short 60 contracts.

Increase beta

From 0.70 to 1.10 on a USD 10m portfolio with USD 200,000 per contract:

$$N^* = (0.70 - 1.10)\frac{10{,}000{,}000}{200{,}000} = -20.$$

Go long 20 contracts.

Residual risk

Beta hedges remove broad market exposure only approximately. Stock selection, sector tilts, changing beta, dividends, liquidity, and basis risk remain.

Section 6

Stack-and-Roll Hedging

1. Hedge Mechanics
2. Basis Risk
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Stack-and-Roll Hedging

- Use nearby liquid contracts when the exposure lasts longer than liquid futures maturities.
- Close the old contract and open a new one as maturity approaches.
- Each roll creates risk because the old and new contracts can move differently.

Risk-Management Lesson

Stack-and-roll can turn price risk into liquidity and funding risk because futures losses must be paid in cash before the physical exposure is realized.

Section 7

Wrap-Up

1. Hedge Mechanics
2. Basis Risk
3. Forward and Futures Pricing
4. Commodity Behavior
5. Stock Index Futures and Beta Hedging
6. Stack-and-Roll Hedging
7. Wrap-Up

What You Should Know After Lecture 2

- Decide whether a hedge should be short or long.
- Compute effective hedge prices using the final basis.
- Explain why cross hedges need an estimated hedge ratio.
- Use futures to hedge or change stock-market beta.
- Derive forward prices from no-arbitrage with income, yield, storage, and convenience yield.
- Explain why futures prices are pricing objects, not automatically unbiased forecasts.

Practice Before Next Class

1. Compute the effective price in a short hedge and in a long hedge.
2. Estimate the minimum-variance hedge ratio from ρ, σ_S, and σ_F.
3. Calculate the number of contracts for a stock-index beta hedge.
4. Price a forward with no income, known income, and known yield.
5. Explain when convenience yield can keep a commodity futures price below a simple cash-and-carry benchmark.

Appendix: Derivation of the Minimum-Variance Hedge Ratio

Start from the hedged change and choose h to minimize its variance:

$$\begin{aligned}
\Delta V &= \Delta S + h(-\Delta F) = \Delta S - h\Delta F, \\
\mathrm{Var}(\Delta V) &= \mathrm{Var}(\Delta S - h\Delta F) \\
&= \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F), \\
\frac{d}{dh}\mathrm{Var}(\Delta V) &= 2h\sigma_F^2 - 2\,\mathrm{Cov}(\Delta S, \Delta F).
\end{aligned}$$

Set the derivative equal to zero and solve:

$$\begin{aligned}
0 &= 2h\sigma_F^2 - 2\,\mathrm{Cov}(\Delta S, \Delta F), \\
h^*\sigma_F^2 &= \mathrm{Cov}(\Delta S, \Delta F), \\
h^* &= \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\mathrm{Var}(\Delta F)}.
\end{aligned}$$

Now use $\mathrm{Cov}(\Delta S, \Delta F) = \rho\sigma_S\sigma_F$ and $\mathrm{Var}(\Delta F) = \sigma_F^2$:

$$h^* = \frac{\rho\sigma_S\sigma_F}{\sigma_F^2} = \rho\frac{\sigma_S}{\sigma_F}.$$

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